

Identity Theft & Spoofing Risk Analysis

Cybersecurity Analytics Report

Prepared Using Python, Pandas, Matplotlib, Streamlit & ReportLab

Generated Automatically

Total Login Records : 5801

Total Alerts : 1340

Suspicious Users : 98

Project Overview

The Identity Theft & Spoofing Risk Analysis system detects suspicious login activities using rule-based cybersecurity analytics. The project focuses on identifying users who exhibit suspicious behaviour through multiple security rules. The implemented detection modules include: • Impossible Travel Detection • Device Spoofing Detection • Odd Hour Login Detection Each detected activity is assigned a LOW, MEDIUM or HIGH risk score which helps security analysts prioritize investigation.

Executive Summary

The system processed **5801** login records belonging to **100** unique users. A total of **1340** suspicious activities were detected. **98** users were classified as HIGH risk. **0** users were classified as MEDIUM risk. **0** users were classified as LOW risk. Overall, **98** users require further security investigation.

Project Statistics

Metric	Value
Total Login Records	5801
Unique Users	100
Total Alerts	1340
Suspicious Users	98
High Risk Users	98
Medium Risk Users	0
Low Risk Users	0

Risk Distribution

Risk Level	Number of Users
HIGH	98

Alert Distribution

Alert Type	Count
IMPOSSIBLE_TRAVEL	1313
ODD_HOURS	15
DEVICE_SPOOFING	12

Top Suspicious Users

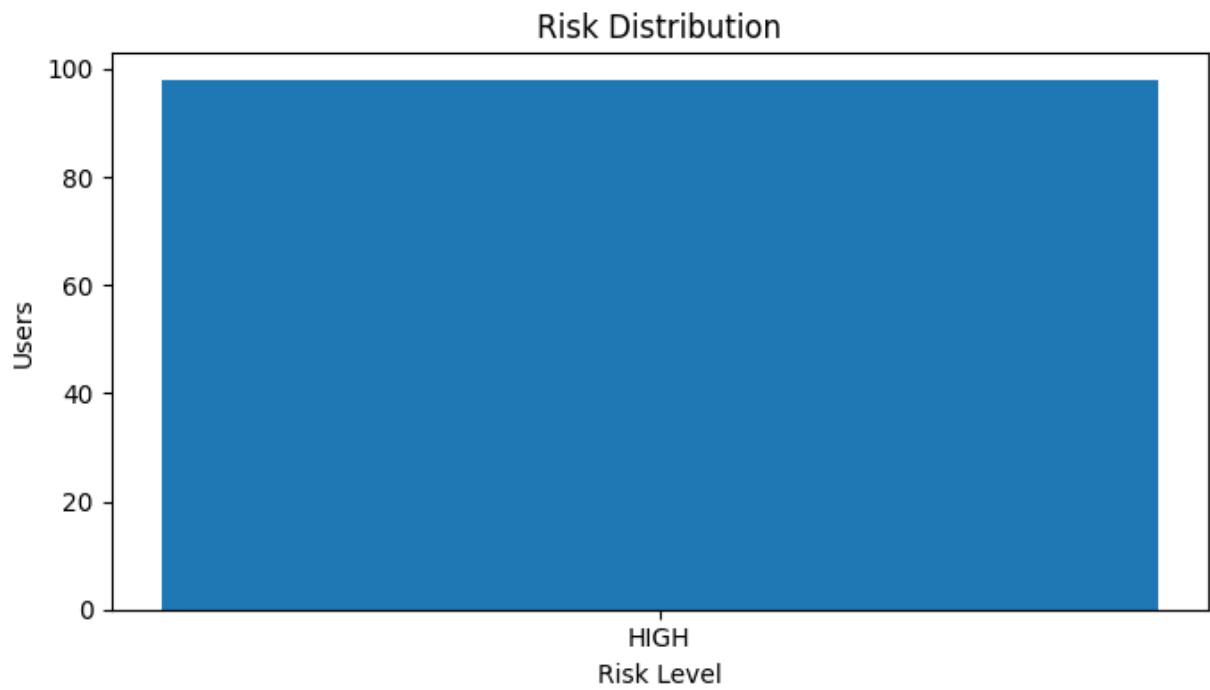
User ID	Total Alerts	Highest Score
USER_0095	34	90
USER_0055	33	90
USER_0063	33	90
USER_0044	32	90
USER_0098	32	90
USER_0021	31	90
USER_0009	30	90
USER_0030	29	90
USER_0038	29	90
USER_0092	29	90

City Distribution

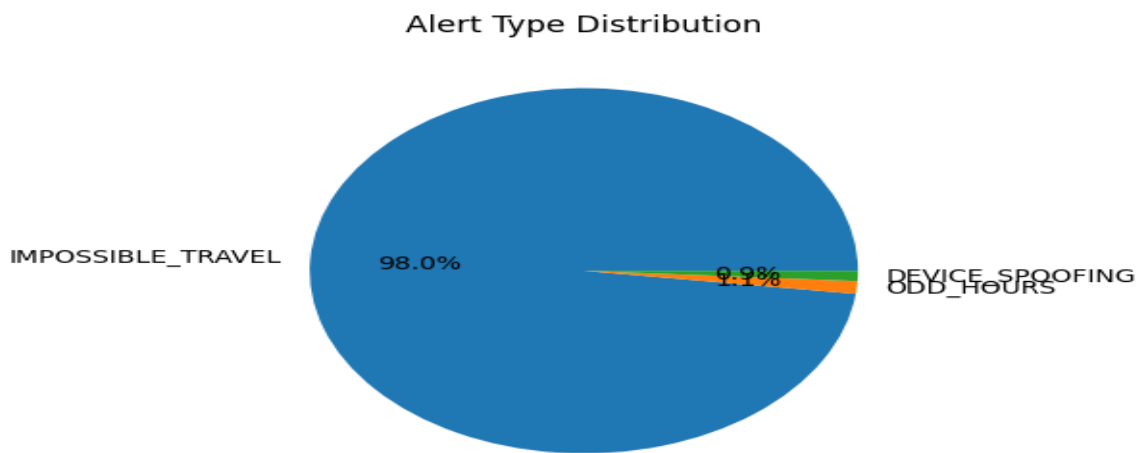
City	Suspicious Logins
Tokyo	158
Sydney	154
Los Angeles	149
Paris	148
Mumbai	145
Singapore	135
New York	132
Dubai	122
Toronto	102
London	95

Visual Analysis

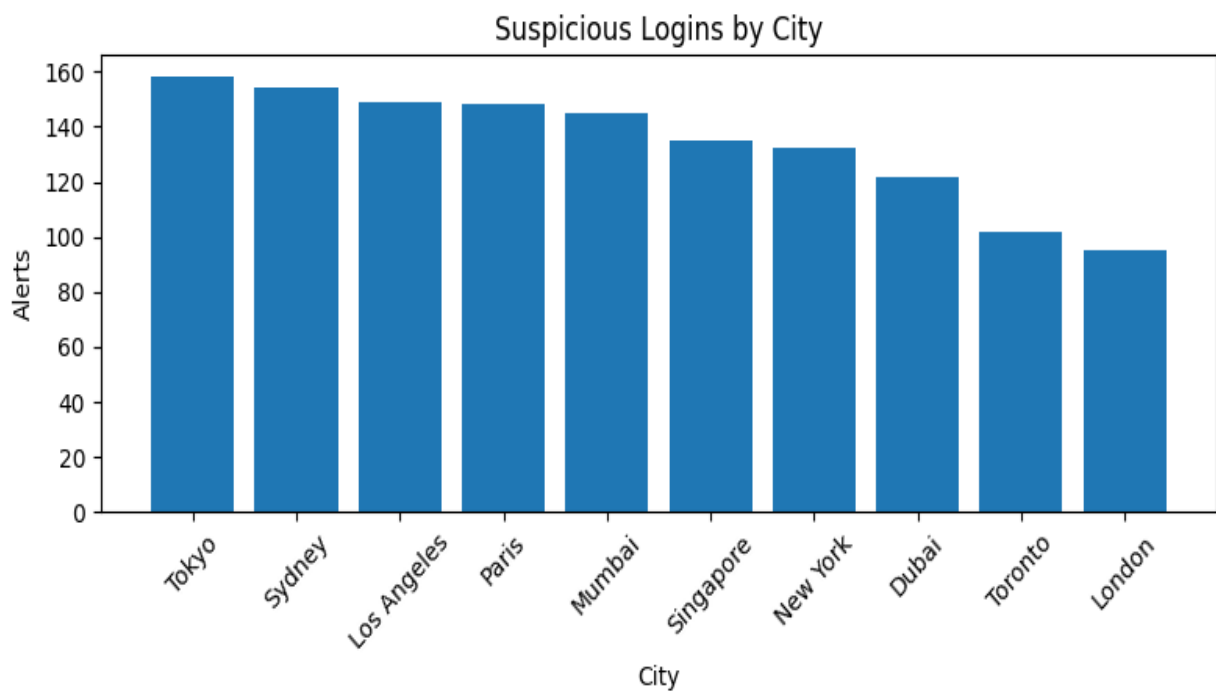
1. Risk Distribution



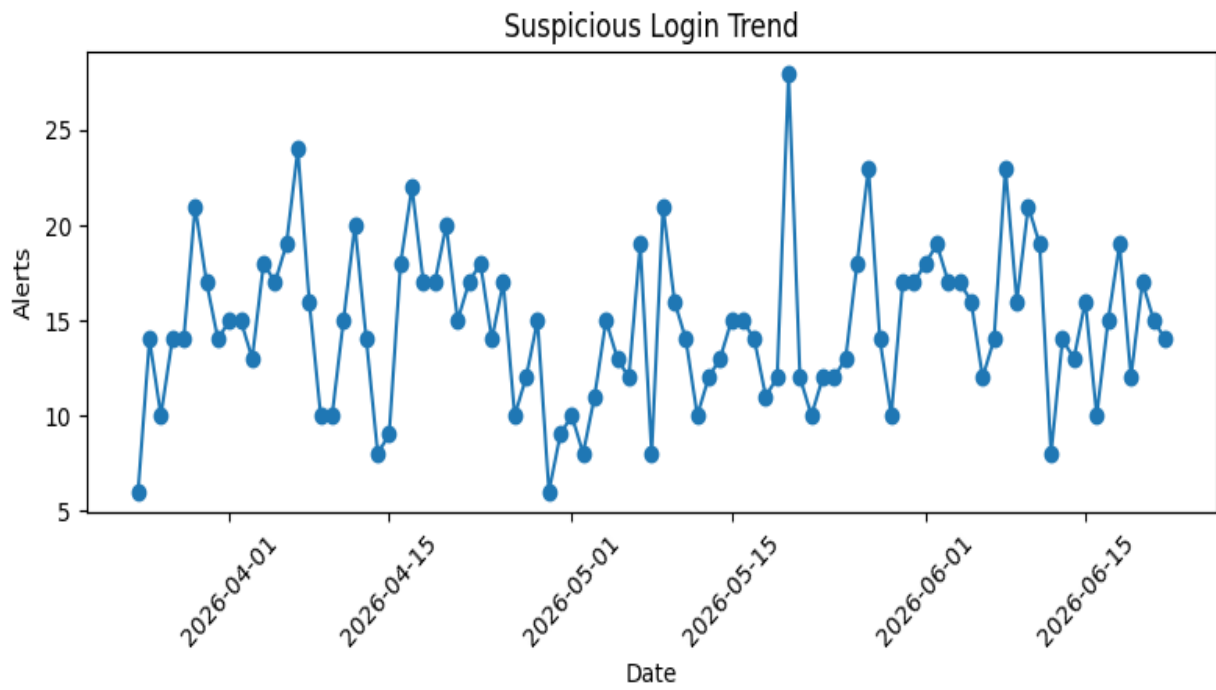
2. Alert Type Distribution



3. City Distribution



4. Suspicious Login Trend



Attack Analysis

The cybersecurity analytics engine processed **5801** login events and generated **1340** security alerts. The most frequently detected attack was **IMPOSSIBLE_TRAVEL** with **1313** occurrences. The city with the highest number of suspicious login activities was **Tokyo** with **158** alerts. User **USER_0095** generated the largest number of alerts (**34**), indicating repeated suspicious activity. The observed attack patterns suggest possible credential theft, identity spoofing, account compromise, or unauthorized login attempts. Early detection of these activities enables security teams to investigate incidents before they escalate into successful attacks.

Security Recommendations

1. Enable Multi-Factor Authentication (MFA)
2. Block impossible travel login attempts automatically.
3. Verify newly detected devices before allowing access.
4. Notify users immediately after suspicious login activity.
5. Continuously monitor HIGH-risk accounts.
6. Enable device fingerprint verification.
7. Apply geo-location validation for login requests.
8. Increase monitoring during unusual login hours.
9. Maintain centralized audit logs for forensic investigations.

Conclusion

The Identity Theft & Spoofing Risk Analysis system successfully analyzed **5801** login records and detected **1340** potentially suspicious login activities. By combining rule-based detection techniques, risk scoring, interactive dashboards, and automated PDF reporting, the system provides security analysts with a simple yet effective method for identifying identity theft and account compromise attempts. This project demonstrates how Python-based cybersecurity analytics can improve organizational security monitoring, reduce manual investigation effort, and support faster incident response.

End of Cybersecurity Analytics Report

Generated automatically using Python, Pandas, Matplotlib and ReportLab.